THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rhombus (not a square) has how many lines of symmetry? 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 0

2. The smallest angle through which a regular pentagon must be rotated to coincide with itself is (degrees): 1

(a) 102 **(b)** 82 **(c)** 62 **(d)** 72

3. The mirror image of the letter A in a vertical mirror to its right looks like: 1

(a) a reversed letter **(b)** a different letter

(c) an upside-down letter **(d)** the same letter

4. A figure has rotational symmetry of order 3. After rotating through 240° it looks the same 2 time(s). The angle of one such turn is (degrees): 1

(a) 110 **(b)** 130 **(c)** 150 **(d)** 120

5. A regular polygon has 7 lines of symmetry. The number of its sides is: 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 9

6. Which of these letters has rotational symmetry of order 2 but NO line of symmetry? 1

(a) Z **(b)** O **(c)** A **(d)** H

7. The figure rotated by 45 degrees about its centre coincides with itself when its order of symmetry is: 1

(a) 7 **(b)** 8 **(c)** 10 **(d)** 9

8. A parallelogram (not a rectangle or rhombus) has how many lines of symmetry? 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 4

9. The smallest angle through which a regular hexagon must be rotated to coincide with itself is (degrees): 1

(a) 60 **(b)** 90 **(c)** 70 **(d)** 50

10. The mirror image of the letter M in a vertical mirror to its right looks like: 1

(a) a different letter **(b)** the same letter

(c) a reversed letter **(d)** an upside-down letter

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Draw (describe) all lines of symmetry of a rectangle (not square). 2

12. Find the angle of rotational symmetry and order for a regular polygon of 5 sides. 2

13. Write the mirror image of the word MAM in a vertical mirror (the reflection reads reversed) and say if the word looks the same. 2

14. Draw (describe) all lines of symmetry of a rhombus (not square). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A regular polygon has rotational symmetry of angle 40°. How many sides does it have, and how many lines of symmetry? 3

16. Give an example (letter or shape) with 3
(i) a line of symmetry but no rotational symmetry (order 1),
(ii) rotational symmetry but no line of symmetry, and explain with A.

17. A point P is 3 units above a horizontal mirror line. Where is its image and how far apart are P and its image? 3

OR

A regular polygon has rotational symmetry of angle 45°. How many sides does it have, and how many lines of symmetry? 3

18. Give an example (letter or shape) with 3
(i) a line of symmetry but no rotational symmetry (order 1),
(ii) rotational symmetry but no line of symmetry, and explain with M.

OR

A point P is 4 units above a horizontal mirror line. Where is its image and how far apart are P and its image? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Study the shape: a regular hexagon. Write 5
(a) number of lines of symmetry,
(b) order of rotational symmetry,
(c) smallest angle of rotation,
(d) the angle after 4 such rotations.

OR

Letters of the word MATH: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

20. Study the shape: a regular octagon. Write 5
(a) number of lines of symmetry,
(b) order of rotational symmetry,
(c) smallest angle of rotation,
(d) the angle after 3 such rotations.

OR

Letters of the word TOOT: find which letters have vertical symmetry, which have horizontal symmetry and which have rotational symmetry of order 2. Is the word itself symmetric about a vertical mirror? 5

END OF QUESTION PAPER — Total: 40 Marks